

LAB_014_Activity

June 4, 2026

1 Activity: Debug Python Code

1.1 Introduction

One of the biggest challenges faced by analysts is ensuring that automated processes run smoothly. Debugging is an important practice that security analysts incorporate in their work to identify errors in code and resolve them so that the code achieves the desired outcome.

Through a series of tasks in this lab, you'll develop and apply your debugging skills in Python.

Tips for completing this lab

As you navigate this lab, keep the following tips in mind:

- **### YOUR CODE HERE ###** indicates where you should write code. Be sure to replace this with your own code before running the code cell.
- Feel free to open the hints for additional guidance as you work on each task.
- To enter your answer to a question, double-click the markdown cell to edit. Be sure to replace the “[Double-click to enter your responses here.]” with your own answer.
- You can save your work manually by clicking File and then Save in the menu bar at the top of the notebook.
- You can download your work locally by clicking File and then Download and then specifying your preferred file format in the menu bar at the top of the notebook.

1.2 Scenario

In your work as a security analyst, you need to apply debugging strategies to ensure your code works properly.

Throughout this lab, you'll work with code that is similar to what you've written before, but now it has some errors that need to be fixed. You'll need to read code cells, run them, identify the errors, and adjust the code to resolve the errors.

1.3 Task 1

The following code cell contains a syntax error. In this task, you'll run the code, identify why the error is occurring, and modify the code to resolve it. (To ensure that it has been resolved, run the code again to check if it now functions properly.)

```
[2]: # For loop that iterates over a range of numbers
      # and displays a message each iteration
```

```
for i in range(10):
    print("Connection cannot be established")
```

```
Connection cannot be established
Connection cannot be established
Connection cannot be established
Connection cannot be established
Connection cannot be established
Connection cannot be established
Connection cannot be established
Connection cannot be established
Connection cannot be established
Connection cannot be established
```

Hint 1

The header of a for loop in Python requires specific punctuation at the end.

Hint 2

The header of a for loop in Python requires a colon (:) at the end.

Question 1 What happens when you run the code before modifying it? How can you fix this?

You see a syntax error so I just added the missing colon and it works now.

1.4 Task 2

In the following code cell, you're provided a list of usernames. There is an issue with the syntax. In this task, you'll run the cell, observe what happens, and modify the code to fix the issue.

```
[4]: # Assign `usernames_list` to a list of usernames
```

```
usernames_list = ["djames", "jpark", "tbailey", "zdutchma", "esmith",
    ↪ "srobinso", "dcoleman", "fbautist"]
```

```
# Display `usernames_list`
```

```
print(usernames_list)
```

```
['djames', 'jpark', 'tbailey', 'zdutchma', 'esmith', 'srobinso', 'dcoleman',
'fbautist']
```

Hint 1

Each element in `usernames_list` is a username and should be a string. In Python, a string should have quotation marks around it.

Hint 2

When creating a list in Python, the elements of the list should be separated with commas. There should be a comma between every two consecutive elements.

Question 2 What happens when you run the code before modifying it? How can you fix it?

There is a syntax error because `usernames_list` was missing a comma and a quotation mark.

1.5 Task 3

In the following code cell, there is a syntax error. Your task is to run the cell, identify what is causing the error, and fix it.

```
[6]: # Display a message in upper case  
  
print("update needed".upper())
```

UPDATE NEEDED

Hint 1

Calling a function in Python requires both opening and closing parentheses.

Hint 2

In the code above, check that each function call has both opening and closing parentheses.

Question 3 What happens when you run the code before modifying it? What is causing the syntax error? How can you fix it?

I see `SyntaxError: unexpected EOF while parsing`. There was a missing parentheses at the end.

1.6 Task 4

In the following code cell, you're provided a `usernames_list`, a `username`, and code that determines whether the username is approved. There are two syntax errors and one exception. Your task is to find them and fix the code. A helpful debugging strategy is to focus on one error at a time and run the code after fixing each one.

```
[10]: # Assign `usernames_list` to a list of usernames that represent approved users  
  
usernames_list = ["djames", "jpark", "tbailey", "zdutchma", "esmith",  
    ↪ "srobinso", "dcoleman", "fbautist"]
```

```

# Assign `username` to a specific username

username = "esmith"

# For loop that iterates over the elements of `usernames_list` and determines
↳ whether each element corresponds to an approved user

for name in usernames_list:

    # Check if `name` matches `username`
    # If it does match, then display a message accordingly

    if name == username:
        print("The user is an approved user")

```

The user is an approved user

Hint 1

In Python, the = assignment operator allows you to assign or reassign a variable to a value, and the == comparison operator allows you to compare one value to another (or the value of one variable to the value of another).

Hint 2

Indentation is important in Python syntax. Check that the indentation inside the for loop and the indentation inside the if statement are correct.

Hint 3

Check that each time a variable is used, it's spelled in the same way it was spelled when it was assigned.

Question 4 What happens when you run the code before modifying it? What is causing the errors? How can you fix it?

There was three errors, but they did not show all at once. There was a missing equal sign, print wasn't indented and usernames_list was misspelled.

1.7 Task 5

In this task, you'll examine the following code and identify the type of error that occurs. Then, you'll adjust the code to fix the error.

```

[15]: # Assign `usernames_list` to a list of usernames

usernames_list = ["elarson", "bmoreno", "tshah", "sgilmore", "eraab"]

# Assign `username` to a specific username

```

```
username = "eraab"

# Determine whether `username` is the final username in `usernames_list`
# If it is, then display a message accordingly

if username == usernames_list[4]:
    print("This username is the final one in the list.")
```

This username is the final one in the list.

Hint 1

Recall that indexing in Python starts at 0.

Hint 2

Identify how many elements there are in the `usernames_list`.

Hint 3

Since indexing in Python starts at 0 and the `usernames_list` contains 5 elements, identify which index value corresponds to the final element in `usernames_list`.

Question 5 What happens when you run the code before modifying it? What type of error is this? How can you fix it?

I was getting an error where it said that the list index was out of range. This is an `IndexError`. To fix it all I had to do was go one number back in the list to put it back in range.

1.8 Task 6

In this task, you'll examine the following code. The code imports a text file into Python, reads its contents, and stores the contents as a list in a variable named `ip_addresses`. It then removes elements from `ip_addresses` if they are in `remove_list`. There are two errors in the code: first a syntax error and then an exception related to a string method. Your goal is to find these errors and fix them.

```
[24]: # Assign `import_file` to the name of the text file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addressess that are no longer allowed to
→access the network

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.
→58.57"]

# With statement that reads in the text file and stores its contents as a list
→in `ip_addresses`
```

```

with open(import_file, "r") as file:
    ip_addresses = file.read()

# Convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# For loop that iterates over the elements in `remove_list`,
# checks if each element is in `ip_addresses`,
# and removes each element that corresponds to an IP address that is no longer
→allowed

for element in remove_list:
    if element in ip_addresses:
        ip_addresses.remove(element)

# Display `ip_addresses` after the removal process

print(ip_addresses)

```

```

['ip_address', '192.168.25.60', '192.168.205.12', '192.168.6.9',
'192.168.52.90', '192.168.90.124', '192.168.186.176', '192.168.133.188',
'192.168.203.198', '192.168.218.219', '192.168.52.37', '192.168.156.224',
'192.168.60.153', '192.168.69.116']

```

Hint 1

A `with` statement in Python requires a colon (`:`) at the end of the header.

Hint 2

The `.split()` method in Python is used on strings to convert them to lists. To call the `.split()` method, place the string you want to split in front of the method call.

Question 6 What happens when you run the code before modifying it? What is causing the errors? How can you fix them?

Before fixing, the code fails due to a missing colon and incorrect use of the `.split()` method.

1.9 Task 7

In this final task, there are three operating systems: OS 1, OS 2, and OS 3. Each operating system needs a security patch by a specific date. The patch date for OS 1 is "March 1st", the patch date for OS 2 is "April 1st", and the patch date for OS 3 is "May 1st".

The following code stores one of these operating systems in a variable named `system`. Then, it uses conditionals to output the patch date for this operating system.

However, this code has logic errors. Your goal is to assign the `system` variable to different values, run the code to examine the output, identify the error, and fix it.

```
[28]: # Assign `system` to a specific operating system as a string

system = "OS 3"

# Assign `patch_schedule` to a list of patch dates in order of operating system

patch_schedule = ["March 1st", "April 1st", "May 1st"]

# Conditional statement that checks which operating system is stored in
→ `system` and displays a message showing the corresponding patch date

if system == "OS 1":
    print("Patch date:", patch_schedule[0])

elif system == "OS 2":
    print("Patch date:", patch_schedule[1])

elif system == "OS 3":
    print("Patch date:", patch_schedule[2])
```

Patch date: May 1st

Hint 1

Recall that indexing in Python starts at 0.

Hint 2

Note that the patch dates in `patch_schedule` are in order of operating system. The first patch date in `patch_schedule` corresponds to OS 1, the second patch date in `patch_schedule` corresponds to OS 2, and so on.

Hint 3

Since indexing in Python starts at 0 and `patch_schedule` is in order of operating system from OS 1 to OS 3, the index value 0 corresponds to the patch date for OS 1, the index value 1 corresponds to the patch date for OS 2, and so on.

Question 7 What happens when you run the code before modifying it? What is causing the logic errors? How can you fix them?

Before the output was giving the wrong patch dates for each OS. The errors were that the conditional statements reference the wrong list indices.

1.10 Conclusion

What are your key takeaways from this lab?

I learned how to identify and fix syntax and logic errors in Python. I practiced reading error messages, correcting missing colons, and properly using string methods like `.split()`. This lab showed how careful debugging ensures code runs correctly and produces accurate results.